

SMS-based Automation System



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Presented here is a system that can be used to remote control devices using a mobile phone. Short messaging service (SMS) is used here to send triggering commands. The main attractive feature of this system is that the user can control the desired device from any part of the world. It works on the global system for mobile communication (GSM). The system is implemented for two devices but can be further developed to control more devices.

Circuit and working

Fig. 1 shows the circuit of SMS-based automation system. It comprises regulator 7805 (IC1), microcontroller P89V51RD2 (IC2), RS232 driver (IC3), GSM module, two BC548 transistors (T1 and T2) and a few discrete components. To provide power supply to the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regu-

Test Points

Test point	Details
TP0	Ground (GND)
TP1	5V
TP2	Unregulated 12V
TP3	High when switch (S1) pressed
TP4	High when message 'fanon' received, low when message 'fanof' received
TP5	High when message 'lgton' received, low when message 'lgtof' received

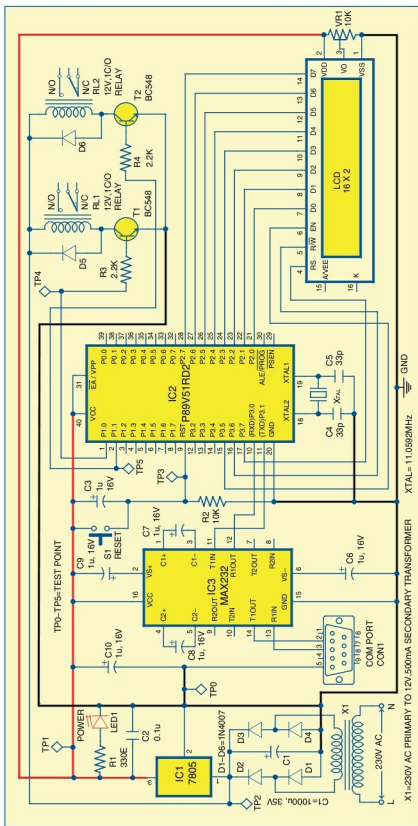


Fig. 1: Circuit of SMS-based automation system